SOIL EROSION AND CONSERVATION · NOTE ER04

Water Erosion — Erosivity, Erodibility, Controlling Factors and Forms of Erosion

Course topics: CAE1.3 L9, SWCE4.5 L4, SWCE4.5 L13, SWCE-501 T7, SWCE-501 T9

Geological and accelerated erosion

	Geological (natural, normal) erosion	Accelerated erosion
When	Under natural vegetation, by water, wind, gravity and glaciers	After vegetation is removed and land cultivated — caused by people
Rate	Very low; balanced by soil formation	Faster than soil formation
Significance	Formed today's soils and their distribution; not serious for farming	What "soil erosion" usually means on cultivated land

Rainfall erosivity

- Water erosion depends on **erosivity** (the rain's potential to erode) and **erodibility** (the soil's susceptibility).
- Erosivity depends on intensity, drop size, drop-size distribution and the kinetic energy of falling rain — the force that detaches particles. Low-intensity rain has more small drops; high-intensity rain more large drops. Drops fall at terminal velocities of 4–10 m/s.

$$KE = \tfrac{1}{2} \cdot M \cdot v^2 \ (v = 8 \text{ m/s} \rightarrow KE = 32\ M)$$
$$KE = 0.119 + 0.0873 \cdot \log_{10} I \text{ for } I \leq 76 \text{ mm/hr}$$
$$KE = 0.283 \text{ for } I > 76 \text{ mm/hr (KE in MJ/ha-mm; I = intensity, mm/hr)}$$

Correction: the notes print KE = 0.119 - 0.873 $\log_{10} I$. The Wischmeier–Smith metric equation is 0.119 ***+ 0.0873*** *$\log_{10} I$ — the minus sign and the misplaced decimal would make the energy negative above about 1.4 mm/hr. Check: at I = 76 mm/hr, 0.119 + 0.0873 × 1.881 = 0.283, matching the cap.*

- **Splash:** drop impact breaks aggregates, splashes soil, seals the surface and raises runoff. A heavy storm on bare soil can splash > 200 tonnes of soil per ha, throwing particles ≈ 60 cm high and ≈ 1.5 m sideways.
- **Erodibility** depends on the soil's physical properties, the land and crop management — especially texture, aggregate stability, shear strength, infiltration and organic matter. Bouyoucos (1935) related it to texture; the dispersion ratio is also used.

$$\text{Bouyoucos erodibility ratio: } E = (\%\ \text{sand} + \%\ \text{silt})/\%\ \text{clay}$$

Runoff erosion: when rainfall exceeds infiltration, surface runoff gains kinetic energy and scours the land; the turbulent flow forms eddies that dislodge and carry particles.

Erosivity indices

- **EI_{30} index:** storm kinetic energy × maximum 30-min intensity (from the recording-gauge chart). Developed for American conditions, so less suited to tropical and sub-tropical rain.

- **KE > 25 index** (Hudson): developed for tropical storms — same procedure, but counts only the energy of rain falling at intensities > 25 mm/hr.

Factors affecting water erosion

$$E = f(C, T, V, S)\ C = \text{climate}, T = \text{topography}, V = \text{vegetation}, S = \text{soil}$$

Some factors can be controlled; others — climate, slope, inherent soil properties — cannot.

Factor	Effect
Climate	Rainfall matters most, then temperature and wind. Drops detach soil and runoff carries it; erosion depends on the agent's detaching and transporting capacity and the soil's detachability and transportability. When rain exceeds infiltration, surface flow forms. Large, intense rains erode severely; frequent rains keep soil wet and raise runoff; evenly spread rain is easier to control. In monsoon Asia, including India, rain is poorly distributed and concentrated in a short season
Topography	Flat land erodes little; sloping land severely. Slope × 4 → velocity × 2; velocity × 2 → erosive capacity × 4, material transported × 32, size of particle moved × 64. A stream carries only limited suspended load and drops it as it slows. Slope length: $E \propto \sqrt{L}$ — longer slopes collect more water, deeper and faster
Vegetation	Dense cover can almost cancel erosion: it intercepts rain (some evaporates) and softens drop impact; slows and spreads runoff; roots bind soil, improve aggregation and create pores that raise infiltration; earthworm and animal channels improve permeability and aeration; transpiration frees soil storage
Soil	Infiltration and resistance to dispersion control erosion. If rain < infiltration there is no runoff and no erosion. Infiltration depends on permeability, surface condition, moisture and structure; bare soil crusts into a thin compact layer, and wet soil infiltrates less. Coarse sand infiltrates fast but erodes in heavy rain; fine clay infiltrates very slowly, sheds runoff on slopes and holds much water. Silt loam, loam and fine sandy loam are the most desirable for minimum erosion

Forms of water erosion

Water erosion removes soil from exposed land by raindrop impact and running water (from rain, snowmelt or ice), detaching particles and carrying them in different forms.

Raindrop (splash) erosion

- Its rate depends on climate (rainfall, temperature), soil (resistance to dispersion, infiltration), topography (slope steepness and length) and plant cover (living plants or residues).
- Drops break aggregates and detach fine particles into a muddy suspension; the fine silt forms a thin sealing film that cuts infiltration and raises runoff. Hard rain sets up a vicious cycle, removing fine fertile material and leaving sand and gravel.
- Large drops can raise sediment-carrying capacity up to 12 times; relative soil loss $\propto V_t^{4.33}$ (V_t = terminal velocity). Terminal velocity depends on drop size, fall height, wind and air resistance — about 4.5 m/s for a 1 mm drop and 9 m/s for a 5 mm drop.
- Wind, slope, surface condition and obstructions (vegetation, mulch) govern splash movement; splash travels farther downhill than uphill.
- Detachability rises and transportability falls with particle size — clay is harder to detach but easier to carry. Detachment strips fine nutrients and speeds erosion.

For reference, Gunn and Kinzer's (1949) measured terminal velocities are about 4.0 m/s for a 1 mm drop and 9.1 m/s for a 5 mm drop.

Form	Description
Sheet erosion	Removes a fairly uniform thin layer by rain and runoff, mainly on gentle slopes. Slow and often unnoticed — the field looks unchanged — yet it strips fertile topsoil uniformly; worst where loose shallow topsoil overlies tight subsoil. Splash is the primary cause: sediment seals pores and water flows off as a thin sheet carrying soil. It has damaged millions of hectares in India
Rill erosion	Running water cuts small, shallow, branching channels from concentrated runoff. No sharp boundary with sheet erosion but more visible; common on loose shallow topsoil; the transition to gullying. Small rills disappear with normal tillage, but neglected rills grow into gullies
Gully erosion	Excessive concentrated runoff cuts channels 30 cm to > 10 m deep — an advanced stage of rills. Runoff concentrates along cattle paths, cart tracks, dead furrows, tillage furrows and small depressions; rills deepen and widen every year into gullies and gully networks that make fertile land uncultivable. A **ravine** is a deep, narrow gorge, larger than a gully
Stream-channel erosion	Scouring of beds and banks by running water in lower stream reaches with near-continuous flow and flat gradients. **Bank cutting**, the commonest form, occurs on the outer bends of meanders through side-flow runoff and undercutting; it is accelerated by removing vegetation, overgrazing and tilling near banks, and depends on velocity and direction of flow, channel depth and width, and soil texture

How gullies grow

- Gully erosion depends on the runoff-producing characteristics and area of the watershed, soil properties, the gully's alignment, size and shape, and the slope of its bed.
- **Waterfall erosion:** water pouring over the gully head undermines the bank, which caves in; the soil is carried off and the head moves upstream. Gullies starting in natural watercourse banks extend into fields, may reach 15–18 m depth or more, and branch into networks.
- **Channel erosion:** concentrated runoff scours depressions, usually into V-shaped gullies that grow longer (faster than wider), deeper and wider; steep upper slopes deepen fastest.
- Waterfall erosion extends the gully headward; channel erosion enlarges its bed and sides.

1. **Formation** — channel scour of the topsoil.
2. **Development** — the head moves upstream; the gully widens and deepens.
3. **Healing** — vegetation starts to grow.
4. **Stabilisation** — stable gradient; vegetative cover spreads. Runoff into the head falls as its drainage area shrinks, and the rest enters along the gully's length.

Class (Tejwani and Dhruva Narayana, 1961)	Depth	Bed width	Side slope
G_1 — very small	≤ 3 m	≤ 18 m	
G_2 — small	≤ 3 m	> 18 m	8–15%
G_3 — medium	3–9 m	≥ 18 m	8–15%
G_4(a) — deep and narrow	3–9 m	< 18 m	
G_4(b) — deep and narrow	> 9 m		Steep or vertical

By cross-section, gullies are V-shaped or U-shaped. (The ICAR eCourse also classes them by size — small < 1 m deep and < 2 ha drainage area, medium 1–5 m and 2–20 ha, large > 5 m and > 20 ha — and

adds trapezoidal gullies, formed where a resistant layer limits deepening.)

Sediment movement and results of erosion

- Sediment moves in **suspension**, by **saltation**, and as **bed load** (surface creep), depending on velocity and turbulence; particle size, cohesion and specific gravity; channel roughness and obstructions. In water, saltation rises only ≈ 1/1000 as high as in air, so it matters little; bed load is 10–54% of total sediment.

1. **Loss of productive soil:** fertile topsoil and organic matter go; fines reach the oceans while coarse material settles on river beds and plains, where it is harmful and agriculturally useless.
2. **Sand deposition on fields:** coarse sediment from hills makes plains unproductive.
3. **Silting of lakes and reservoirs:** less storage and a shorter useful life — high sedimentation is reported in major Indian reservoirs.
4. **Silting of drainage and irrigation channels:** less depth and capacity → overflow, flooding, crop and structural damage, poor drainage and waterlogging.
5. **Lower water table:** more runoff means less infiltration and recharge, so wells yield less.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Wischmeier, W. H. and Smith, D. D. (1978), Predicting Rainfall Erosion Losses, USDA Agriculture Handbook 537. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Doubling flow velocity increases the size of particle that can be moved by about

(a) 4 times (b) 8 times
(c) 32 times (d) 64 times

Q2. Doubling velocity increases the quantity of material transported by about

(a) 4 times (b) 16 times
(c) 32 times (d) 64 times

Q3. Rainfall kinetic energy (MJ/ha-mm) for intensities above 76 mm/hr is taken as

(a) 0.119 (b) 0.283
(c) 0.873 (d) 1.0

Q4. The KE > 25 erosivity index was developed for

(a) temperate storms (b) tropical storms
(c) snowmelt (d) wind

Q5. Bouyoucos' erodibility ratio is

(a) % clay/% sand (b) (% sand + % silt)/% clay
(c) % silt/% clay (d) % OM/% clay

Q6. A raindrop splash on bare soil can throw particles about

(a) 5 cm high (b) 60 cm high and 1.5 m sideways
(c) 3 m high (d) 10 m sideways

Q7. The erosion form that is slow, uniform and often unnoticed is

(a) rill (b) gully
(c) sheet (d) stream-bank

Q8. A gully more than 9 m deep with steep sides is class

(a) G_1 (b) G_2

(c) G_3 (d) G_4(b)

Q9. Bed load forms about ___ of total sediment in channels

(a) 1–5% (b) 10–54%

(c) 60–80% (d) > 90%

Q10. Headward extension of a gully is due to

(a) channel erosion (b) waterfall erosion

(c) sheet erosion (d) saltation

Answers

Q1 — (d) 64 times

Q2 — (c) 32 times

Q3 — (b) 0.283

Q4 — (b) tropical storms

Q5 — (b) (% sand + % silt)/% clay

Q6 — (b) 60 cm high and 1.5 m sideways

Q7 — (c) sheet

Q8 — (d) G_4(b)

Q9 — (b) 10–54%

Q10 — (b) waterfall erosion